MIL-A-40032E
30 May 1986
SUPERSEDING
MIL-A-40032D
24 June 1971

MILITARY SPECIFICATION

APRON, MEAT CUTTER'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of protective, metal mesh, meat cutter's apron.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
V-T-276	- Thread, Cotton
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-B-654	- Brazing Alloys, Silver
DDD-L-20	- Label: For Clothing, Equipage and Tentage, (General Use)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-R-3390 - Rings, Dee
MIL-H-9890 - Hardware, Individual Load Carrying Equipment, and Hardware, Miscellaneous
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-S-43770 - Snap Hooks, Cast Body, Open Eye, Sliding Bolt Closure

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-798 - Apron, Meat Cutter's: Assembly and Details
5-11-828 - Apron, Meat Cutter's; Strap and Webbing Details and Sections

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. Unless otherwise specified, the following document of the issue in effect on the date of invitation for bids or request for proposal forms a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. The specified materials shall conform to applicable documents and requirements specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for his meat cutter's aprons provided that the completed items conform to all provisions of this document. Recycled material may be used for nontextile components (see 6.4).

3.2.1 Brass, sheet or strip. Brass, sheet or strip, shall conform to the requirements of composition 4, any temper, of QQ-B-613.

3.2.2 Eyelets and washers. Eyelets and washers shall be fabricated of brass and finished as specified in 3.4.

3.2.2.1 Eyelets. Eyelets shall be rolled flange type with an outside diameter of not less than 0.20 inch, a flange diameter of not less than 0.34 inch, and a metal thickness of not less than 0.011 inch.

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3.2.2.2 Washers. Washers shall be round flat type with an outside diameter of not less than 0.40 inch, a thickness of not less than 0.018 inch, and an inside diameter to accommodate the eyelets specified in 3.2.2.1.

* 3.2.3 Solder. Solder shall conform to the requirements of grade VII of QQ-B-654.

3.2.4 Thread. Thread shall be ticket No. 12, 4 ply and shall conform to the requirements for type IA1 or type IB1 of V-T-276.

3.2.5 Cotton webbing. Cotton webbing shall conform to the requirements for type IIb, class 1, 1 inch wide, 0.96 ounce per linear yard of MIL-W-530.

3.2.6 Keeper clamps. Keeper clamps shall conform to the requirements for type IX, size 1 inch with loop, class A or class B of MIL-H-9890, except that the finish shall be as specified in 3.4.

* 3.2.7 Snaphooks. Snaphooks shall conform to the requirements for type 1, class 1 or class 2, of MIL-S-43770.

3.2.8 Dee rings. Dee rings shall conform to the requirements for class 1 or class 2, size 1 inch by 7/8 inch of MIL-R-3390, except that the finish shall be as specified in 3.4.

* 3.2.9 Labels. In addition to the "US" marking required by Drawing 5-11-828, each apron shall have an identification label conforming to type VI, class 1 of DDD-L-20.

3.3 Construction. The construction shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.3.1 Metal mesh. The metal mesh shall be formed from the ring and square, 4 prong stampings of the brass and assembled as shown on Drawing 5-11-798. The prongs of the square stampings shall be soldered.

3.3.2 Eyelets and washers. The eyelets shall be inserted through the washers, the webbing, and the metal mesh and set as shown on Drawing 5-11-798. The eyelet holes through the cotton webbing shall be smaller than the outside diameter of the eyelet shank so that the eyelet shank must be forced through the webbing. The diameter of the eyelet holes in the metal mesh shall be not more than 0.005 inch larger than the outside diameter of the eyelet shank.

* 3.3.3 Stitching. All stitching shall conform to type 301 of FED-STD-751. Six to eight stitches per inch shall be maintained in all stitching operations.

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* 3.3.3.1 Automatic stitching. Automatic stitching machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and that at least three tying, overlapping, or backstitches are used to secure the ends of stitching.

* 3.3.3.2 Type 301 stitching. Ends of all stitching except W-W shall be backstitched or overstitched a minimum of 1 inch except where ends are turned under in a hem or held down by other stitching. On box-X stitching, all ends shall be backstitched or overstitched a minimum of 1/2 inch. Thread tension shall be maintained so there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the sewn material. The lock shall be embedded in the sewn materials.

* 3.3.3.2.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for box-X stitching) back of the end of the stitching.1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area (1/2 inch for box-X stitching), continue over the defective area, and continue a minimum of 1 inch (1/2 inch for box-X stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner.1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

* 3.3.3.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.4 Finish. The finish on all metal components shall be decorative nickel plate, nominal 0.0001 inch thickness.

* 3.5 Workmanship. The finished apron shall conform to the quality of the product established by this document. The occurrence of defects shall not exceed the applicable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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* 4.4.1.1 In-process inspection. Inspection shall be made of the following processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the affected items and lot in process.

a. Soldering (see 3.3.1).
b. Eyelet holes in bib, belt, and bottom webbing (see 3.3.2).
c. Diameter of eyelet holes in metal mesh (see 3.3.2).

The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

* 4.4.2 End item visual examination. The aprons shall be examined for the defects listed in table I. The lot size shall be expressed in units of aprons. The sample unit shall be one apron. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor combined) defects.

TABLE I. Classification of visual defects

Examine	Defect	Classification	
		Major	Minor
Soldering	Cold solder joints (joints not completely sealed)	X	
	Not clean and free of embedded foreign matter		X
	Excess solder and flux not removed		X
Finish: Nickel plate (metal parts)	Not plated	X	
	Any visible blister, pit, porosity, or indication of burning	X	
	Any nodule or excessive edge buildup		X
	Plating not adherent	X	
Design	Any characteristic not in accordance with specified requirements	X	

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TABLE I. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship, general (applicable to all components and assemblies)	Any component missing	X	
	Any component misplaced; operation omitted or not properly performed	X	
	Two or more rings not engaging prongs from appropriate stampings as specified except at the sides of the apron	X	
	Any ring not engaging prongs from four separate stampings as specified except at the sides of the apron		X
	Any prongs on one stamping not joined together by solder		X
	Two prongs on two or more stampings not joined together by solder	X	
	Any sharp burr, edge, or protrusion	X	
	Any component or assembly malformed		X
	Attachment of any webbing border piece or strap to apron not accomplished as specified	X	
	Any cut, hole, or tear in webbing components:		
	- where dee ring is attached or where keeper clamp or spring snap is attached	X	
	- in any area other than that specified above		X
Stitching (webbing)	Not type specified		X
	Any row of stitching missing	X	
	Any open seam 1/4 inch or more	X	
	Any open seam less than 1/4 inch		X
	One stitch per inch less than the minimum specified		X
	More than one stitch per inch less than the minimum specified	X	
	More stitches per inch than maximum specified, damaging webbing	X	

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TABLE I. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Stitching (webbing) - continued	More stitches per inch than maximum specified, but not damaging webbing		X
	Loose stitch tension		X
	Tight stitch tension		X
Label	Missing, incomplete, incorrect, not legible, not specified type or size, not in proper location, or not accomplished in the specified manner		X

* 4.4.3 End item dimensional examination. The aprons shall be examined for conformance to the dimensions specified on the drawings listed in section 2. Any dimension not within specified tolerances shall constitute a defect. The lot size shall be expressed in units of aprons. The sample unit shall be one apron. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of bundles per container is more or less than required

* 4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each apron shall be laid out flat, inside up, with straps placed upon the apron and rolled to form an oval shaped bundle measuring approximately 18 1/2 by 3 by 2 inches. Each rolled apron shall then be completely and compactly wrapped in minimum 60-pound kraft paper conforming to A-A-203, or placed in a kraft bag made of the same material. The wrap shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. Aprons shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twenty aprons, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, one in length, four in width, and five in depth. The inside dimensions of the shipping container shall be approximately 18 1/2 inches in length, 12 inches in width, and 10 inches in depth. The approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twenty aprons, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a liner conforming to type CF, variety DW, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, one in length, four in width, and five in depth. The inside dimensions of the shipping container shall be approximately 18 1/2 inches in length, 12 inches in width, and 10 inches in depth. The approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Aprons, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), aprons, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with

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bonding means K and L or film bonding means O or P. Pallet pattern shall be Number 8 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The apron is intended to give protection to the chest, abdomen, groin, and front of thighs of meat cutters. It is used in central meat cutting plants of the Armed Forces.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B shipments (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material for nontextile components when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-40032E	APRON, MEAT CUTTER'S

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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